

# Mathematical and Computational Methods

## Unit 8: Differentiation II: applications

Unit 7 built the derivative and the rules for computing it. This unit puts that machinery to work. We begin with the geometry it controls—the *tangent* and *normal* lines—and with four theorems that underpin everything which follows: the intermediate-value, extreme-value and mean-value theorems, and the rule for differentiating an inverse, together with *l'Hôpital's rule* for awkward limits. We then use the first and second derivatives to locate and classify the turning points of a curve and to sketch it accurately, and finally to solve *optimisation* problems—closing with *gradient descent*, the numerical idea that turns the derivative into an engine for minimisation across the sciences and data science.

**Where this fits.** Everything rests on Unit 7: the derivative and its three readings, the rules, higher derivatives, and the continuity it requires. The mean-value theorem is what licenses the increasing/decreasing test of Section 2, and the extreme-value theorem is what licenses the endpoint check in the optimisation of Section 3. The curve-sketching skills feed directly into the area and volume problems of Units 9–10, and the optimisation and gradient-descent material is foundational for the data-science and physics students alike.

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## Learning objectives

By the end of this unit you will be able to:

- ▶ Find tangent and normal lines.
- ▶ State the key theorems (IVT, EVT, MVT, inverse-function) and apply l'Hôpital's rule.
- ▶ Solve equations numerically by Newton's method.
- ▶ Locate and classify stationary points with the first- and second-derivative tests.
- ▶ Sketch curves of polynomial and rational functions using calculus.
- ▶ Solve optimisation problems and explain gradient descent.

## 1 Tangents, normals and the key theorems

### 1.1 Tangent and normal lines

The geometric reading of Unit 7 says  $f'(a)$  is the slope of the curve at  $(a, f(a))$ . Two lines through that point are determined by it.

**Definition 1.1 (Tangent and normal lines).** At the point  $(a, f(a))$  on a differentiable curve  $y = f(x)$ , the *tangent line* is

$$y = f(a) + f'(a)(x - a),$$

and the *normal line* is the perpendicular through the same point. When  $f'(a) \neq 0$  the normal has slope  $-1/f'(a)$ , so

$$y = f(a) - \frac{1}{f'(a)}(x - a).$$

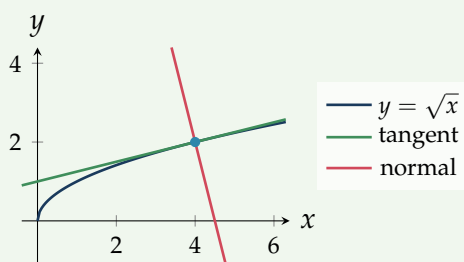
If  $f'(a) = 0$  the tangent is horizontal and the normal is the vertical line  $x = a$ .

The normal slope is  $-1/f'(a)$  because perpendicular lines have slopes whose product is  $-1$ . The normal direction is the one in which a surface pushes back on a body resting against it—the *normal force* of mechanics—which is where the name comes from. (A particle released from circular motion, by contrast, flies off along the *tangent*, not the normal.)

**Example 1.1 (Tangent and normal to  $y = \sqrt{x}$ ).** In Unit 7 we found the tangent to  $y = \sqrt{x}$  at  $x = 4$ : there  $f(4) = 2$  and  $f'(4) = \frac{1}{4}$ , giving the tangent  $y = \frac{1}{4}x + 1$ . The normal at the same point has slope  $-1/f'(4) = -4$ , so

$$y - 2 = -4(x - 4) \quad \implies \quad y = -4x + 18.$$

Tangent and normal meet at right angles at  $(4, 2)$ , as their slopes  $\frac{1}{4}$  and  $-4$  confirm. With equal scales on the two axes, that right angle is visible directly.



## 1.2 Two theorems for continuous functions

The next two theorems cost nothing but continuity, yet they guarantee the existence of things we will need: solutions of equations, and maxima and minima.

**Theorem 1.1 (Intermediate value theorem).** If  $f$  is continuous on  $[a, b]$  and  $N$  is any value strictly between  $f(a)$  and  $f(b)$ , then  $f(c) = N$  for some  $c$  in  $(a, b)$ .

In words, a continuous function takes every value between any two it attains—its graph cannot jump over a level. The most useful case is  $N = 0$ : if  $f$  is continuous and changes sign on  $[a, b]$ , it has a root in between.

**Example 1.2 (Locating a root).** Let  $f(x) = x^3 - x - 1$ . It is continuous, with  $f(1) = -1 < 0$  and  $f(2) = 5 > 0$ . By the intermediate value theorem there is a root in  $(1, 2)$ . This is also the reasoning behind the fact that every odd-degree polynomial has at least one real root: its end behaviour is governed by the leading term (Unit 1), so its values run from  $-\infty$  to  $+\infty$ , or the reverse if the leading coefficient is negative, and either way it must cross zero.

**Theorem 1.2 (Extreme value theorem).** If  $f$  is continuous on a closed, bounded interval  $[a, b]$ , then  $f$  attains an absolute maximum and an absolute minimum on  $[a, b]$ : there are points  $c, d \in [a, b]$  with  $f(d) \leq f(x) \leq f(c)$  for all  $x \in [a, b]$ .

*Note.* Both hypotheses are essential. On the open interval  $(0, 1)$  the function  $f(x) = x$  attains neither a maximum nor a minimum, and on  $(0, 1]$  the function  $1/x$  has no maximum (it runs off to  $\infty$  near 0). The theorem is what guarantees that the optimisation problems of Section 3 actually have a best answer, and that we may find it by checking finitely many candidates.

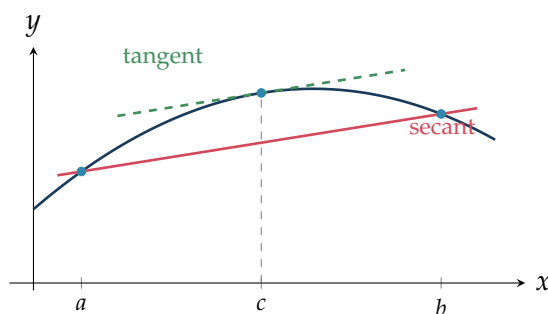
## 1.3 The Mean Value Theorem

Where the previous two theorems used only continuity, the mean value theorem uses differentiability, and it is the workhorse behind the curve-analysis of Section 2.

**Theorem 1.3 (Mean value theorem).** If  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then there is a point  $c \in (a, b)$  with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Geometrically: somewhere strictly between  $a$  and  $b$  the tangent is parallel to the secant joining the endpoints—the instantaneous rate of change equals the average rate of change at least once.



The single most useful consequence concerns the *sign* of the derivative. If  $f'$  is positive throughout an interval, then for any two points the right-hand side above is positive, so  $f$  increases; and likewise for the other signs.

**Corollary 1.1 (Monotonicity from the derivative).** On an interval where  $f$  is differentiable:

$$f' > 0 \Rightarrow f \text{ strictly increasing, } f' < 0 \Rightarrow f \text{ strictly decreasing, } f' \equiv 0 \Rightarrow f \text{ constant.}$$

This is the precise statement behind the intuition “positive slope means going uphill,” and it is the tool we use in Section 2 to read a curve’s shape off its derivative.

## 1.4 The derivative of an inverse function

Unit 1 built inverse functions by reflecting a graph in the line  $y = x$ . That reflection swaps the roles of “rise” and “run,” so it should turn a slope  $m$  into  $1/m$ —which is exactly what happens to the derivative.

**Theorem 1.4 (Derivative of an inverse).** Let  $f$  be differentiable and one-to-one, and write  $a = f^{-1}(b)$ . If  $f'(a) \neq 0$ , then  $f^{-1}$  is differentiable at  $b$  with

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}.$$

The rise/run picture is the intuition; the chain rule of Unit 7 makes it exact. Since  $f^{-1}$  undoes  $f$ , the two compose to the identity,  $f(f^{-1}(x)) = x$ . Differentiating both sides and applying the chain rule on the left gives

$$f'(f^{-1}(x)) \cdot (f^{-1})'(x) = 1 \quad \Rightarrow \quad (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))},$$

which is the formula above. This also explains the hypothesis  $f' \neq 0$ : it is exactly what stops the right-hand side dividing by zero (geometrically, a horizontal tangent of  $f$  reflects to a vertical tangent of  $f^{-1}$ , where the slope is undefined).

**Example 1.3 (Recovering the derivative of  $\ln$ ).** The natural logarithm is the inverse of  $f(x) = e^x$ , for which  $f' = e^x$ . With  $f^{-1}(x) = \ln x$ ,

$$(\ln)'(x) = \frac{1}{f'(\ln x)} = \frac{1}{e^{\ln x}} = \frac{1}{x},$$

the standard derivative of Unit 7. The same method (or, equivalently, implicit differentiation) gives the inverse-trigonometric derivatives

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} \quad (|x| < 1), \quad \frac{d}{dx} \arctan x = \frac{1}{1+x^2},$$

completing the catalogue of standard derivatives begun in Unit 7. The restriction on  $\arcsin$  is real: at  $x = \pm 1$  the denominator vanishes and the graph of  $\arcsin$  meets the ends of its domain vertically, with no finite derivative.

## 1.5 l'Hôpital's rule

Limits of the form  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$  are *indeterminate*: the answer depends on *how* numerator and denominator approach their limits. Differentiation resolves them.

**Theorem 1.5 (l'Hôpital's rule).** Suppose  $f$  and  $g$  are differentiable near  $a$  with  $g' \neq 0$  there, and that  $\frac{f(x)}{g(x)}$  has the indeterminate form  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$  as  $x \rightarrow a$ . Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided the right-hand limit exists (or is  $\pm\infty$ ). The point  $a$  may be finite or  $\pm\infty$ .

**Example 1.4 (Two limits, one repeated).** For the  $\frac{0}{0}$  form

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\sin x}{2x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\cos x}{2} = \frac{1}{2},$$

applying the rule twice (the first application is still  $\frac{0}{0}$ , so we check and differentiate again). For the  $\frac{\infty}{\infty}$  form,

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0,$$

which says  $\ln x$  grows more slowly than  $x$ .

*Note.* l'Hôpital applies *only* to indeterminate forms. Differentiating top and bottom of a limit that is not  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$  gives the wrong answer: always confirm the form first. Other indeterminate forms ( $0 \cdot \infty$ ,  $\infty - \infty$ ,  $1^\infty$ , ...) can usually be rearranged into one of these two.

## 1.6 Newton's method

The intermediate value theorem *locates* a root—it tells us a continuous, sign-changing  $f$  has a zero in  $(1, 2)$ —but it does not say *where*. To pin the root down we put the tangent line to work. A tangent is the curve's best straight-line proxy near the point of contact, so if our current estimate is close to a root, the tangent there crosses the axis very close to where the curve does. Following the tangent to the axis turns a rough estimate into a sharper one; repeating the step drives it home.

Concretely, the tangent to  $y = f(x)$  at the current estimate  $x_n$  is

$$y = f(x_n) + f'(x_n)(x - x_n).$$

Setting  $y = 0$  and solving for  $x$  gives the point where this tangent meets the axis, which we take as the next estimate  $x_{n+1}$ :

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n) \quad \implies \quad \boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}, \quad f'(x_n) \neq 0.$$

This is *Newton's method* (or the Newton–Raphson method): start from a guess  $x_0$  and iterate.

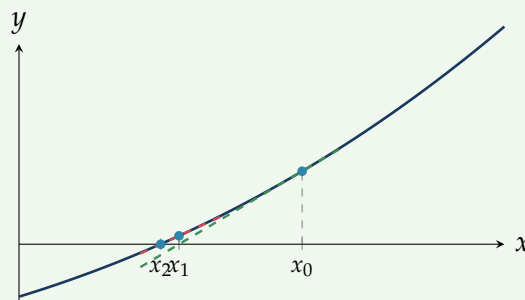
**Example 1.5 (Newton's method on  $x^3 - x - 1$ ).** We return to the root of  $f(x) = x^3 - x - 1$  that the intermediate value theorem placed in  $(1, 2)$ . Here  $f'(x) = 3x^2 - 1$ , so the iteration is

$$x_{n+1} = x_n - \frac{x_n^3 - x_n - 1}{3x_n^2 - 1}.$$

Starting from  $x_0 = 1.5$ :

| $n$ | $x_n$    | $f(x_n)$ |
|-----|----------|----------|
| 0   | 1.500000 | 0.875000 |
| 1   | 1.347826 | 0.100682 |
| 2   | 1.325200 | 0.002058 |
| 3   | 1.324718 | 0.000001 |
| 4   | 1.324718 | 0.000000 |

The estimates settle on  $x \approx 1.324718$ , the real root to six decimal places, in three or four steps. Notice the number of correct digits roughly *doubles* each step ( $0.15 \rightarrow 0.02 \rightarrow 0.0005 \rightarrow \dots$  in the error): this rapid, “quadratic” convergence is what makes the method so useful. The geometry is shown below—each step slides down the tangent to the axis.



*Note (When Newton works—and when it does not).* Newton's method is fast but not foolproof. Started near a simple root (one where  $f' \neq 0$ ) it converges rapidly, but a poor starting guess can send it off to a different root or away altogether, and if  $f'(x_n)$  is near zero the step  $f(x_n)/f'(x_n)$  becomes huge and the iteration can jump wildly. The intermediate value theorem offers the slow-but-safe complement: it is guaranteed to trap a root once a sign change is found, where Newton trades that guarantee for speed. The same “step using the derivative” idea returns in Section 3 as gradient descent—there to minimise rather than to solve  $f = 0$ .

## 2 Critical points and curve sketching

### 2.1 Critical points and the first-derivative test

A high or low point of a graph has a horizontal tangent, so the search for the peaks and troughs of a function starts by setting its derivative to zero.

**Definition 2.1 (Critical and stationary points).** A *critical point* of  $f$  is a value  $x = c$  in its domain where  $f'(c) = 0$  or  $f'(c)$  fails to exist. A point where  $f'(c) = 0$  is also called a *stationary point*. The corresponding point  $(c, f(c))$  is a candidate for a local maximum or minimum.

**Theorem 2.1 (Fermat's theorem).** If  $f$  has a local maximum or minimum at an interior point  $c$  of its domain and  $f'(c)$  exists, then  $f'(c) = 0$ . Consequently an extremum can occur only at a critical point or at an endpoint—which is why checking the critical points and the endpoints suffices.

By the monotonicity corollary,  $f$  increases where  $f' > 0$  and decreases where  $f' < 0$ , so the behaviour of a stationary point is decided by how the sign of  $f'$  changes across it.

**Theorem 2.2 (First-derivative test).** Let  $c$  be a stationary point of  $f$ . As  $x$  increases through  $c$ :

$$f' : + \rightarrow - \Rightarrow \text{local maximum}, \quad f' : - \rightarrow + \Rightarrow \text{local minimum},$$

and if  $f'$  keeps the *same strict sign* on both sides,  $c$  is neither (the curve levels off and continues the same way—a horizontal inflection). Strict is the word: for  $f(x) = (\text{ReLU}(x))^2$  the derivative is 0 to the left of 0 and positive to the right, so it never changes sign, yet  $x = 0$  is a global minimum.

## 2.2 Concavity and the second-derivative test

The second derivative records how the slope is changing, which is the same as the way the curve bends.

**Definition 2.2 (Concavity and inflection).** On an interval,  $f$  is *concave up* (holds water) where  $f'' > 0$  and *concave down* where  $f'' < 0$ . A point where the concavity changes—so  $f''$  changes sign—is an *inflection point*.

At a stationary point the concavity decides the type directly, which is usually quicker than tracking the sign of  $f'$  on both sides.

**Theorem 2.3 (Second-derivative test).** Suppose  $f'(c) = 0$ . Then

$$f''(c) > 0 \Rightarrow \text{local minimum at } c, \quad f''(c) < 0 \Rightarrow \text{local maximum at } c.$$

If  $f''(c) = 0$  the test is inconclusive and one falls back on the first-derivative test.

The test is just the first-derivative test in disguise. Suppose  $f'(c) = 0$  and  $f''(c) > 0$ . Then  $f''$  is the derivative of  $f'$ , so  $f''(c) > 0$  says  $f'$  is *increasing* as  $x$  passes through  $c$ ; since  $f'(c) = 0$ , the slope  $f'$  therefore runs from negative to positive across  $c$ . By the first-derivative test that is a local minimum. The case  $f''(c) < 0$  is the mirror image, with  $f'$  decreasing through zero, hence a maximum. When  $f''(c) = 0$  this argument gives no information— $f'$  need not change sign—which is precisely why the test is then inconclusive: compare  $f(x) = x^4$  (a minimum at 0) with  $f(x) = x^3$  (no extremum), both of which have  $f'(0) = f''(0) = 0$ .

**Example 2.1 (Classifying stationary points).** For  $f(x) = x^3 - 3x$ ,  $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$ , so the stationary points are  $x = \pm 1$ . With  $f''(x) = 6x$ ,

$$f''(1) = 6 > 0 \Rightarrow \text{local minimum at } (1, -2),$$

$$f''(-1) = -6 < 0 \Rightarrow \text{local maximum at } (-1, 2).$$

The second-derivative test settles both in one line each.

*Remark (The sigmoid, revisited).* Unit 7 found  $\sigma'(x) = \sigma(x)(1 - \sigma(x))$  for the logistic function. Since  $0 < \sigma < 1$ , this is positive for every  $x$ , so by the monotonicity corollary  $\sigma$  is strictly increasing everywhere—it never turns back, which is why it makes a good probability. Differentiating again gives  $\sigma'' = \sigma'(1 - 2\sigma)$ , which vanishes only at  $\sigma = \frac{1}{2}$ , i.e.  $x = 0$ : the sigmoid has a single inflection there, switching from concave up to concave down.

## 2.3 Curve sketching

Together,  $f$ ,  $f'$  and  $f''$  give enough information for an accurate sketch without plotting many points. A systematic pass collects:

1. **Domain and symmetry** — where  $f$  is defined; even ( $f(-x) = f(x)$ ) or odd ( $f(-x) = -f(x)$ ) symmetry from Unit 1.
2. **Intercepts** —  $f(0)$  for the  $y$ -axis; solve  $f(x) = 0$  for the  $x$ -axis.
3. **Asymptotes** — for rational functions, vertical asymptotes at the zeros of the denominator that survive cancellation (a zero shared with the numerator may be a removable hole instead, as  $(x^2 - 1)/(x - 1) = x + 1$  is at  $x = 1$ ), and horizontal asymptotes from the limits at infinity of Unit 7.
4. **Increase/decrease and extrema** — the sign of  $f'$  and the stationary points it locates.
5. **Concavity and inflections** — the sign of  $f''$ .

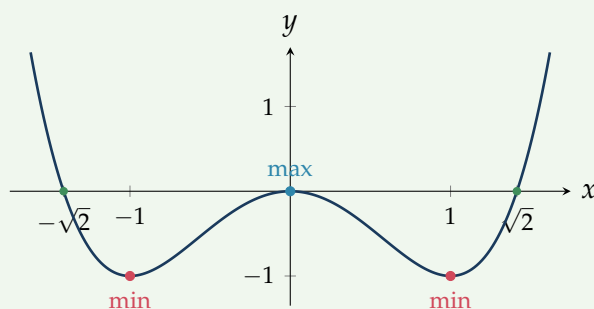
**Example 2.2 (A full sketch:  $f(x) = x^4 - 2x^2$ ).**

- *Domain and symmetry.* Defined on all of  $\mathbb{R}$ ; even, since  $f(-x) = f(x)$ , so the graph is symmetric about the  $y$ -axis.
- *Intercepts.*  $f(0) = 0$ , and  $f(x) = x^2(x^2 - 2) = 0$  gives  $x = 0$  and  $x = \pm\sqrt{2}$ .
- *Increase/decrease.*  $f'(x) = 4x^3 - 4x = 4x(x-1)(x+1)$ , with stationary points  $x = -1, 0, 1$ . The sign of  $f'$  runs  $-, +, -, +$  across these, so  $f$  decreases on  $(-\infty, -1)$ , increases on  $(-1, 0)$ , decreases on  $(0, 1)$  and increases on  $(1, \infty)$ .
- *Classification.*  $f''(x) = 12x^2 - 4$ . Then  $f''(\pm 1) = 8 > 0$ , giving local minima at  $(\pm 1, -1)$ , and  $f''(0) = -4 < 0$ , giving a local maximum at  $(0, 0)$ .
- *Concavity.*  $f'' = 0$  at  $x = \pm\frac{1}{\sqrt{3}}$ , with  $f'' > 0$  outside and  $f'' < 0$  between: the curve is concave up for  $|x| > \frac{1}{\sqrt{3}}$  and concave down between, with inflection points at  $x = \pm\frac{1}{\sqrt{3}}$ .

The resulting “W” shape is drawn below, with the two minima, the central maximum



and the  $x$ -intercepts marked.



A rational function adds the one feature the polynomial above lacks—asymptotes—and this is where the limits of Unit 7 earn their keep.

**Example 2.3** (A rational sketch:  $f(x) = \frac{x^2}{x^2 - 1}$ ).

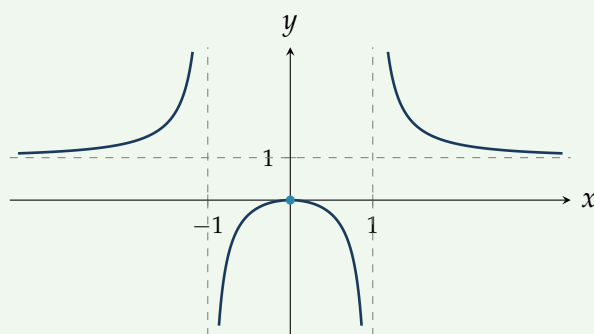
- *Domain and symmetry.* The denominator  $x^2 - 1 = (x - 1)(x + 1)$  vanishes at  $x = \pm 1$ , so the domain is  $\mathbb{R} \setminus \{-1, 1\}$ ; the function is even.
- *Intercepts.*  $f(0) = 0$  is the only intercept.
- *Asymptotes.* Near  $x = \pm 1$  the denominator  $\rightarrow 0$  while the numerator does not, so  $x = \pm 1$  are *vertical* asymptotes. For the behaviour at infinity, divide through by  $x^2$ :

$$\lim_{x \rightarrow \pm\infty} \frac{x^2}{x^2 - 1} = \lim_{x \rightarrow \pm\infty} \frac{1}{1 - 1/x^2} = 1,$$

so  $y = 1$  is a *horizontal* asymptote.

- *Increase/decrease.* By the quotient rule  $f'(x) = \frac{-2x}{(x^2 - 1)^2}$ , which is zero only at  $x = 0$ . Since the denominator is positive,  $f' > 0$  for  $x < 0$  and  $f' < 0$  for  $x > 0$ : the only stationary point is a local maximum at  $(0, 0)$ .
- *Concavity.* Differentiating again,  $f''(x) = \frac{2(3x^2 + 1)}{(x^2 - 1)^3}$ , whose sign is that of  $(x^2 - 1)^3$ : the curve is concave *up* on the outer branches ( $|x| > 1$ ) and concave *down* in the middle ( $|x| < 1$ ), with no inflection (the sign change occurs only across the asymptotes).

Between the vertical asymptotes the curve opens *downward* to  $-\infty$ , with its peak the local maximum at the origin  $(0, 0)$ ; on the outer branches it descends from  $+\infty$  towards the horizontal asymptote  $y = 1$ , staying above it. This gives the three-piece shape below.



### 3 Optimisation

#### 3.1 Solving optimisation problems

Optimisation asks for the largest or smallest value of some quantity. The calculus is the easy part; the skill is translating a description into a function of *one* variable and identifying the interval on which it is to be optimised.

*Note (A procedure for optimisation).*

1. Name the quantity  $Q$  to be optimised and the variables it depends on.
2. Use the constraints to write  $Q$  as a function of a *single* variable, and state the domain (the physically meaningful interval).
3. Find the critical points: solve  $Q' = 0$ , and add any interior point of the domain where  $Q'$  fails to exist (a corner or cusp is a candidate too— $Q(x) = x^{2/3}$  has its minimum at just such a point).
4. Identify the optimum among the critical points and, on a closed interval, the *endpoints*; classify with the second-derivative test where convenient.

The endpoint check in step 4 is not optional: by the extreme value theorem a continuous function on a closed interval attains its extremes *somewhere*, but that somewhere may be a boundary rather than a stationary point. Checking the critical points and the endpoints is therefore guaranteed to find the answer.

**Example 3.1 (The most economical can).** A closed cylindrical can must hold a volume of  $1000 \text{ cm}^3$ . What dimensions use the least metal?

The quantity to minimise is the surface area  $A$  of a cylinder of radius  $r$  and height  $h$ ,

$$A = \underbrace{2\pi r^2}_{\text{two ends}} + \underbrace{2\pi rh}_{\text{side}}.$$

The volume constraint  $\pi r^2 h = 1000$  fixes  $h = \frac{1000}{\pi r^2}$ , which removes  $h$  and leaves a function of  $r$  alone:

$$A(r) = 2\pi r^2 + 2\pi r \cdot \frac{1000}{\pi r^2} = 2\pi r^2 + \frac{2000}{r}, \quad r > 0.$$

Setting the derivative to zero,

$$A'(r) = 4\pi r - \frac{2000}{r^2} = 0 \quad \implies \quad r^3 = \frac{500}{\pi}, \quad r = \left(\frac{500}{\pi}\right)^{1/3} \approx 5.42 \text{ cm}.$$

Since  $A''(r) = 4\pi + \frac{4000}{r^3} > 0$  for all  $r > 0$ , this critical point is a minimum—and as  $r \rightarrow 0$  or  $r \rightarrow \infty$  the area grows without bound, so it is the global minimum. The corresponding height is

$$h = \frac{1000}{\pi r^2} = 2r,$$

so the most economical can is exactly as tall as it is wide. (The tidy result  $h = 2r$  follows on substituting  $r^3 = 500/\pi$  into  $h = 1000/(\pi r^2)$ .)

### 3.2 Gradient descent

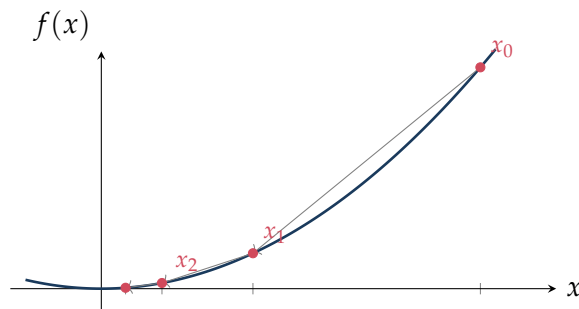
Solving  $f'(x) = 0$  by hand works for the textbook functions above, but in data science a quantity to be minimised—an *error* or *loss*  $L$ —may depend on thousands of parameters through a formula far too tangled to solve exactly. We instead minimise *numerically*, and the derivative shows the way.

Recall the third reading of the derivative from Unit 7:  $f'(x)$  is a *sensitivity*, the direction and rate in which  $f$  responds to a nudge in  $x$ . If  $f'(x) > 0$  then  $f$  rises to the right, so to *decrease*  $f$  we should step to the *left*—against the derivative. This is the whole idea.

**Note (Gradient descent).** To minimise  $f$ , start from a guess  $x_0$  and repeatedly step against the derivative,

$$x_{n+1} = x_n - \eta f'(x_n),$$

where the *learning rate*  $\eta > 0$  sets the step size. Provided  $\eta$  is small enough—made precise below—the iterates move downhill towards a stationary point, where  $f' = 0$  and the steps stop. Too large an  $\eta$  overshoots and climbs instead: on  $f(x) = x^2$  with  $\eta = 1.5$  the step  $x \mapsto -2x$  sends  $1 \mapsto -2 \mapsto 4$ , away from the minimum.



**Example 3.2 (Descending a parabola).** Minimise  $f(x) = x^2$ , for which  $f'(x) = 2x$ , starting at  $x_0 = 2.5$  with  $\eta = 0.3$ . The update is  $x_{n+1} = x_n - 0.3(2x_n) = 0.4x_n$ , so

$$x_0 = 2.5, \quad x_1 = 1.0, \quad x_2 = 0.4, \quad x_3 = 0.16, \quad \dots \rightarrow 0,$$

homing in on the minimum at  $x = 0$  (shown above). The steps shrink as the slope flattens—the method naturally slows as it nears the bottom.

How small is “too small” and how large is “too large”? A quadratic model makes the threshold exact. Consider gradient descent applied to  $f(x) = \frac{1}{2}k(x - x^*)^2$  with  $k > 0$ , a parabola with its minimum at  $x^*$ ; then  $f'(x) = k(x - x^*)$  and the update  $x_{n+1} = x_n - \eta f'(x_n)$  becomes

$$x_{n+1} - x^* = (x_n - x^*) - \eta k(x_n - x^*) = (1 - \eta k)(x_n - x^*).$$

So the *error*  $e_n = x_n - x^*$  obeys  $e_{n+1} = (1 - \eta k)e_n$ : a geometric sequence with ratio  $1 - \eta k$ . It decays to zero exactly when the ratio has magnitude below one,

$$|1 - \eta k| < 1 \quad \Longleftrightarrow \quad 0 < \eta < \frac{2}{k},$$

and it is fastest at  $\eta = 1/k$ , where the ratio is 0 and a single step lands exactly on  $x^*$ . For  $\eta > 2/k$  the ratio exceeds one in magnitude, so the error grows and the iterates oscillate *outward* and diverge—the overshoot made precise. (For the parabola  $f(x) = x^2$  above,  $k = 2$ , so the safe range is  $0 < \eta < 1$  and  $\eta = 0.3$  sits comfortably inside it.)

*Remark (Choosing the step, and what it costs).* The learning rate must therefore be chosen with care: too small and convergence crawls; too large—beyond  $2/k$  for the model above—and the iterates overshoot the minimum and diverge. Gradient descent also finds only a *local* minimum—the one in whose valley it starts—so the starting point matters. With many parameters the single derivative becomes a *vector* of derivatives (the *gradient*), and the step is taken against it; the principle is unchanged.

## Summary

| Concept                | Key idea                                                                                                                                                           |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tangent & normal       | At $(a, f(a))$ : tangent $y = f(a) + f'(a)(x - a)$ ; normal slope $-1/f'(a)$ (vertical if $f'(a) = 0$ ).                                                           |
| Intermediate value     | Continuous $f$ on $[a, b]$ hits every value between $f(a)$ and $f(b)$ ; a sign change forces a root.                                                               |
| Extreme value          | Continuous $f$ on a closed bounded $[a, b]$ attains an absolute max and min—so optimisation has an answer.                                                         |
| Mean value             | Some $c$ has $f'(c) = \frac{f(b) - f(a)}{b - a}$ ; hence $f' > 0 \Rightarrow$ increasing, $f' < 0 \Rightarrow$ decreasing, $f' \equiv 0 \Rightarrow$ constant.     |
| Inverse derivative     | $(f^{-1})'(b) = 1/f'(f^{-1}(b))$ ; gives $(\ln)' = 1/x$ , $\arctan' = 1/(1 + x^2)$ , $\arcsin' = 1/\sqrt{1 - x^2}$ .                                               |
| l'Hôpital              | For $\frac{0}{0}$ or $\frac{\infty}{\infty}$ , $\lim \frac{f}{g} = \lim \frac{f'}{g'}$ ; only for indeterminate forms.                                             |
| Critical points        | $f'(c) = 0$ or undefined; candidates for local extrema.                                                                                                            |
| First-derivative test  | $f' : + \rightarrow -$ max; $- \rightarrow +$ min; same strict sign both sides, neither.                                                                           |
| Concavity              | $f'' > 0$ concave up, $f'' < 0$ concave down; sign change of $f''$ is an inflection.                                                                               |
| Second-derivative test | $f'(c) = 0$ with $f''(c) > 0$ min, $f''(c) < 0$ max, $f''(c) = 0$ inconclusive.                                                                                    |
| Curve sketching        | Domain/symmetry, intercepts, asymptotes (vertical at denominator zeros that do not cancel, horizontal from limits at infinity), $f'$ (extrema), $f''$ (concavity). |
| Optimisation           | Reduce to one variable, find the critical points ( $Q' = 0$ or $Q'$ undefined), classify, check endpoints (extreme value theorem).                                 |
| Gradient descent       | $x_{n+1} = x_n - \eta f'(x_n)$ : step against the derivative towards a minimum; $\eta$ is the learning rate.                                                       |

*Next unit: a change of direction—integration, the reverse of differentiation. We meet the definite integral as an area, the Fundamental Theorem of Calculus linking it to the derivative, and the method of substitution.*

Companion material: a separate *Exercise Sheet 8* (with solutions) develops the practice problems for this unit.